

Math 1452 Fall 2012 Final Exam

Grading Suggestions and Solutions Packet

I have annotated the Key with the point values I plan to use in red. There are 19 different parts, so that works out to 5 points per part, except 1a (4 points), the integrals in problem 3 (6 points each), and problem 6 (7 points). This distributes the point values among the chapters as follows:

Chapter 6	24 points
Chapter 7	29 points
Chapter 8	32 points
Chapter 9	15 points

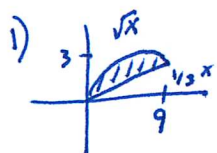
These are of course only suggestions, but we've been requested to make the grading of the finals as uniform as possible.

Calculators are not allowed on this exam. Work all questions completely. Show all work as described in class.
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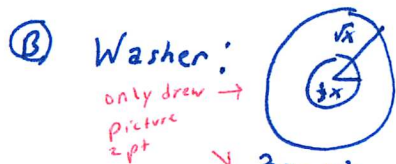
1. Consider the region bounded by the curves $y = \frac{1}{3}x$ and $y = \sqrt{x}$. Set up integrals to find
 - 4 (a) The area of this region.
 - 5, 5 (b) The volume of the solid generated by rotating this region about the x -axis using **both** washers **and** shells.
 - 5 (c) The volume of the solid generated by rotating this region about the vertical line $x = -4$ using any method.
- 5 2. A cylindrical tank has radius of 3 ft and height of 10 ft. It is filled with liquid weighing 76 lb/ft³. Set up an integral to find the work done in pumping out the liquid.
3. Evaluate the following integrals.

6 (a) $\int \frac{2x - 9}{x(x^2 + 9)} dx$	6 (b) $\int e^{2x} \cos(3x) dx$
6 (c) $\int x^2 \sin^2(x^3 - 4) dx$	6 (d) $\int \frac{1}{\sqrt{x^2 - 9}} dx$
- 5 4. Evaluate $\int_0^3 \frac{5}{(x - 2)^2} dx$.
5. Indicate if the following series converge or diverge. You must identify all the tests you use and show all the work needed to apply them.

5 (a) $\sum_{k=1}^{\infty} \frac{4k^2 - 3}{k^3 + 2k}$	5 (b) $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}$
5 (c) $\sum_{k=1}^{\infty} k e^{-k}$	5 (d) $\sum_{k=2}^{\infty} \frac{(2k)!}{3^k}$
- 7 6. Find the interval and radius of convergence of the power series $\sum_{k=0}^{\infty} \frac{2}{3^k k} (x - 5)^k$.
- 5 7. Give the Maclaurin series for $f(x) = 3x^5 \sin(2x)$.
8. If $\mathbf{u} = \langle 1, 0, 3 \rangle$ and $\mathbf{v} = \langle 2, 1, 0 \rangle$, find
 - 5 (a) $\mathbf{u} - 3\mathbf{v}$
 - 5 (b) $\mathbf{u} \times \mathbf{v}$
 - 5 (c) The angle between $\mathbf{u}$ and $\mathbf{v}$



1) (A) Area = $\int_0^9 \sqrt{x} - \frac{1}{3}x \, dx = \int_0^3 3y - y^2 \, dy$



only draw picture 2 pt

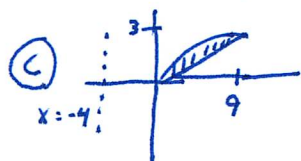
$\int_0^9 \pi (\sqrt{x})^2 - \pi (\frac{1}{3}x)^2 \, dx$

used y variable or y-axis -3

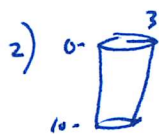
Shell

$\int_0^3 2\pi y (3y - y^2) \, dy$

used x variable or x-axis -3



$\int_0^9 2\pi (x+4) (\sqrt{x} - \frac{1}{3}x) \, dx \quad \text{or} \quad \int_0^3 \pi (3y+4)^2 - \pi (y^2+4)^2 \, dy$



$\int_0^{10} x (76) \pi (3)^2 \, dx \quad \text{or} \quad \int_0^{10} (10-x) (76) \pi (3)^2 \, dx$

3) (A) $\int \frac{2x-9}{x(x^2+9)} \, dx = \int \frac{A}{x} + \frac{Bx+C}{x^2+9} \, dx = \int \frac{-1}{x} + \frac{x+2}{x^2+9} \, dx = \int \frac{-1}{x} + \frac{x}{x^2+9} + \frac{2}{x^2+9} \, dx$

Stopped here give 1 pt

$A(x^2+9) + x(Bx+C) = 2x-9$

$A+B=0 \quad 9A=-9 \quad C=2$

$B=1 \quad A=-1$

$-\ln|x| + \frac{1}{2} \ln|x^2+9| + \frac{2}{3} \tan^{-1} \frac{x}{3} + C$

(B) $\int e^{2x} \cos(3x) \, dx = \frac{1}{3} e^{2x} \sin(3x) - \frac{2}{3} \left(\frac{-1}{3} e^{2x} \cos(3x) - \frac{2}{3} \int e^{2x} \cos(3x) \, dx \right)$

Parts twice

$= \frac{9}{13} \left(\frac{1}{3} e^{2x} \sin(3x) + \frac{2}{9} e^{2x} \cos(3x) \right) + C$

-1 pt each forgotten + C

(C) $\int x^2 \sin^2(x^3-4) \, dx = \frac{1}{3} \int \sin^2 u \, du = \frac{1}{6} \int 1 - \cos(2u) \, du = \frac{1}{6} \left((x^3-4) - \frac{1}{2} \sin(2(x^3-4)) \right) + C$

$u = x^3-4 \quad du = 3x^2 \, dx$

(D) $\int \frac{1}{\sqrt{x^2-9}} \, dx = \int \frac{1}{3 \tanh \theta} 3 \sec \theta \tanh \theta \, d\theta = \int \sec \theta \, d\theta = \ln |\sec \theta + \tanh \theta| + C$

$x = 3 \sec \theta$

$dx = 3 \sec \theta \tanh \theta$



$\ln \left| \frac{x}{3} + \frac{\sqrt{x^2-9}}{3} \right| + C$

4) $\int_0^3 \frac{5}{(x-2)^2} \, dx = \lim_{N \rightarrow 2^-} \int_0^N \frac{5}{(x-2)^2} \, dx + \lim_{N \rightarrow 2^+} \int_N^3 \frac{5}{(x-2)^2} \, dx$

$\lim_{N \rightarrow 2^-} \left. \frac{-5}{x-2} \right|_0^N + \lim_{N \rightarrow 2^+} \left. \frac{-5}{x-2} \right|_N^3$

(both) parts Diverge (actually, we need only check one since it diverges)
So original integral Diverges

Did not treat as improper -4

5) (A) $\sum \frac{4k^2-3}{k^3+2k}$ Limit Comp. Test $\lim_{k \rightarrow \infty} \frac{\frac{4k^2-3}{k^3+2k}}{\frac{1}{k}} = 4$ Diverges b/c $\sum \frac{1}{k}$ Div.

(B) $\sum \frac{(-1)^k}{\sqrt{k}}$ Alt. Series Test $\lim_{k \rightarrow \infty} \frac{1}{\sqrt{k}} = 0$ Converges (Conditionally)

(C) $\sum k e^{-k}$ Ratio Test $\lim_{k \rightarrow \infty} \frac{(k+1)e^{-k-1}}{k e^{-k}} = \frac{1}{e} < 1$ Converges

Root Test $\lim_{k \rightarrow \infty} \sqrt[k]{k e^{-k}} = e^{-1} < 1$ converges

(D) $\sum \frac{(2k)!}{3^k}$ Divergence Test $\lim_{k \rightarrow \infty} \frac{(2k)!}{3^k} = \infty$ Diverges

Ratio Test $\lim_{k \rightarrow \infty} \frac{(2k+2)!}{3^{k+1}} \cdot \frac{3^k}{(2k)!} = \lim_{k \rightarrow \infty} \frac{(2k+1)(2k+2)}{3} = \infty$ Diverges

6) $\sum \frac{2}{3^k k} (x-5)^k$

$\lim_{k \rightarrow \infty} \frac{2|x-5|^{k+1}}{3^{k+1}(k+1)} \cdot \frac{3^k k}{2|x-5|^k} = \frac{|x-5|}{3} < 1 \Rightarrow |x-5| < 3$
Radius = 3

check $x=8 \leq \frac{2}{3^k k} 3^k = \sum \frac{2}{k}$ Div.

Did not check -3

Interval is $[2, 8)$

$x=2 \leq \frac{2}{3^k k} (-3)^k = \sum \frac{2}{k} (-1)^k$ Conv.

7) $3x^5 \sin(2x) = 3x^5 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} (2x)^{2k+1} = \sum_{k=0}^{\infty} 3 \frac{(-1)^k}{(2k+1)!} 2^{2k+1} x^{2k+6}$

8) (A) $u = \langle 1, 9, 3 \rangle$ $v = \langle 2, 1, 0 \rangle$

$u - 3v = \langle -5, -3, 3 \rangle$

(B) $u \times v = \begin{vmatrix} i & j & k \\ 1 & 9 & 3 \\ 2 & 1 & 0 \end{vmatrix} = \langle -3, 6, 1 \rangle$

1 pt per sign error, 1 pt per wrong number

~~7 pt for wrong computation~~
~~1 pt for wrong computation~~
but only 1 pt for it

(C) $u \cdot v = 2 = \sqrt{10} \sqrt{5} \cos \theta$ So $\theta = \cos^{-1} \left(\frac{2}{\sqrt{10} \sqrt{5}} \right)$

$\|u\| = \sqrt{10}$

$\|v\| = \sqrt{5}$